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Communicating over multiple radio access technologies (RATS)

Abstract

A user equipment (UE) is configured to communicate, using a first radio access technology (RAT) having a first paired spectrum, wherein the first paired spectrum has a first frequency for downlink and a second frequency for uplink. The UE is further configured to transmit, using the first radio access technology (RAT), a service request message including information associated with at least one UE service. The UE is configured to receive, based on the transmitted service request message, a message with information to setup a bearer for a second RAT, wherein the second RAT is a different RAT than the first RAT. The UE is configured to receive multimedia data using the second RAT and simultaneously receive data using the first RAT.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/219,499, filed Jul. 7, 2023, which will issue as U.S. Pat. No. 12,069,511 on Aug. 20, 2024, which is a continuation of U.S. patent application Ser. No. 17/946,768, filed Sep. 16, 2022, which issued as U.S. Pat. No. 11,700,544 on Jul. 11, 2023, which is a continuation of U.S. patent application Ser. No. 17/403,448, filed Aug. 16, 2021, which issued as U.S. Pat. No. 11,470,503 on Oct. 11, 2022, which is a continuation of U.S. patent application Ser. No. 16/842,150, filed Apr. 7, 2020, which issued as U.S. Pat. No. 11,096,085 on Aug. 17, 2021, which is a continuation of U.S. patent application Ser. No. 16/113,446, filed Aug. 27, 2018, now abandoned, which is a continuation of U.S. patent application Ser. No. 15/131,264, filed Apr. 18, 2016, which issued as U.S. Pat. No. 10,064,092 on Aug. 28, 2018, which is a continuation of U.S. patent application Ser. No. 14/016,559, filed Sep. 3, 2013, which issued as U.S. Pat. No. 9,319,930 on Apr. 19, 2016, which is a continuation of U.S. patent application Ser. No. 11/502,929, filed Aug. 11, 2006, which issued as U.S. Pat. No. 8,532,653 on Sep. 10, 2013, which is incorporated by reference as if fully set forth.

BACKGROUND OF INVENTION
(1) Mobile operators worldwide have launched streaming real-time services (such as mobile TV and radio) over their existing 3G Universal Mobile Telecommunications System (UMTS)

networks. However, the existing UMTS air-interface and overall network architecture are not adequate to deliver high quality, bandwidth-demanding multimedia content, such as television for a large number of users. Consequently, the 3GPP standards consortium identifies optimizations in the UTRAN and the core network system architecture that will allow the deployment of broadcasting-type applications over a UMTS air-interface and core network.

(2) The add-on framework to the UMTS system architecture in a family of 3GPP specifications is called Multimedia Multicasting/Broadcasting Service (MBMS). The MBMS framework identifies the modifications needed in the UMTS Radio Access Network (RAN) and describes service aspects, such as security and charging in a set of 3GPP specifications (see, for example, references [1], [2], [4], [8]). 3GPP2 has defined a similar service in a family of specifications called Broadcast Multicast Service (BCMCS).

(3) MBMS/BCMCS represents a point-to-multipoint service architecture defined in an end-to-end manner within the family of 3GPP/2 specifications that allow a 3G operator to deploy a broadcasting/multicasting service. For example, Mobile TV is deployed over its allocated spectrum by upgrading the existing network with the relevant 3GPP2 MBMS/BCMCS specifications.

(4) MBMS/BCMCS multimedia streaming traffic and the associated uplink/downlink signaling are conventionally transmitted from the same network on the same frequency band. Because MBMS is intended to serve a large user population, the high volume of broadcast/multicast traffic and the number of processing nodes along its path impose a significant performance strain on the existing core network and UTRA elements of the UMTS network. For example, the core network mobility anchor (e.g., SGSN in 3GPP UMTS), and the radio network controller (e.g., RNC in 3GPP UMTS) must process and transport the "normal" point-to-point traffic generated by the different packet and circuit-switched applications. Therefore, a way to reduce the strain on the network is desired.

BRIEF SUMMARY OF THE INVENTION

(5) Embodiments of the invention provide methods and apparatus to alleviate potential capacity problems in the radio access and core networks operating MBMS by separating the control and user plane of the MBMS traffic (a "one tunnel" approach) across two different Radio Access Technologies (RAT) of the 3GPP family of RATs, and their equivalent Core Network transport path.

(6) Overload traffic conditions for multimedia services may be reduced by transmitting the traffic related to broadcasting multimedia over a first wireless network (defined by one RAT), and the associated control information over a second wireless network (defined by a substantially different RAT). The two networks may operate on substantially different frequency bands. Thus, the wireless terminal may receive a plurality of signals originating from a plurality networks.

(7) A mobile operator may have both a paired and an unpaired spectrum. In the unpaired spectrum, the mobile radio technology may use Time Division Duplexing (TDD). In the paired spectrum, the mobile radio technology may use Frequency Division Duplexing (FDD). The multicast traffic is highly asymmetric. Thus, the multicast traffic is adequately matched to the unpaired spectrum, in which the ratio of downlink to uplink traffic may be adjusted to reflect the traffic demand. Moreover, the ratio may be adjusted such that the entire unpaired spectrum is dedicated to downlink. In this way, maximum utilization of the available spectrum is achieved. Under this scenario, the return channel is no longer present in the unpaired spectrum, and the user relies on the paired spectrum to provide the return channel.

(8) More particularly, embodiments of the invention separate the paths of the multicasting/broadcasting traffic across multiple access and core networks into Control Plane (CP) and User Plane (UP) multicasting/broadcasting traffic. UP multicasting/broadcasting traffic is the actual multimedia content delivered in the downlink. CP is the supplemental traffic that is associated with the UP multicasting/broadcasting traffic. Both the UP and CP are relevant to network-specific signaling procedures and "higher-layer" application-related signaling that applies to the multicasting/broadcasting user service.

(9) The coupling between a first network and a second network by a home gateway allows for a direct tunnel transmission from a broadcast/multicast service center to a user equipment. User equipment (UE) may transmit control uplink signals via the first network. Simultaneously, the UE receives downlink control signals via the first network. The broadcast multimedia service is received at the user equipment via a second network in response to the exchange of uplink and downlink signals between the user equipment and the first network through the home gateway.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an exemplary embodiment of logical architecture of a network with Multimedia Multicasting/Broadcasting Service (MBMS) capabilities.

(2) FIG. **2** illustrates an exemplary embodiment of the signaling flow to establish a direct user plane tunnel for the delivery of multicasting traffic.

(3) FIG. **3** illustrates an exemplary embodiment of a UMTS network with Multimedia Multicasting/Broadcasting Service (MBMS) capabilities.

(4) FIG. **4** illustrates an exemplary embodiment of the signaling flow for MBMS multicast delivery mode signaling procedures, using "one tunnel" approach and downlink-only TD-CDMA radio interface.

(5) FIG. **5** illustrates an exemplary embodiment of the detailed steps involved in MBMS context provisioning in RAN with direct user plane.

(6) FIG. **6** illustrates an exemplary embodiment of a computing system capable of carrying out the functionality of the various embodiments.

DETAILED DESCRIPTION OF THE INVENTION

(7) FIG. **1** illustrates an exemplary embodiment of the logical architecture of a network with a point-to-multipoint service center (e.g., an MBMS service center). A first network is defined by RAT1, and a second network is defined by RAT2. Each RAT is implemented by a radio controller (e.g., RAT1 radio controller **106** and RAT2 radio controller **108**). The radio controller (RC) is the control element in the radio access network (RAN) responsible for controlling the base stations **104** of a specific RAT. The RC carries out radio resource and mobility management functions. This is the point in the network where encryption may be done before user data is sent to and from the mobile UE **102**.

(8) The UE **102** communicates with RAT1 and RAT2 via base stations **104** coupled to the RAT1 RC **106** and RAT2 RC **108**. The UE wireless terminal **102** includes a first receiver, a second receiver, and a transmitter. The UE **102** may receive signals simultaneously from RAT1 and from RAT2. In other words, the first receiver may receive signals over two different networks, which may operate in different frequency bands.

(9) The mobility anchor **110** is the gateway between the RAN and the core network. Mobility anchoring is performed when the UE **102** is moving across a RC of the same or different RATs.

(10) The home gateway **112** is a router that serves as a gateway between mobile networks and packet data networks. It is the ingress/egress point for the data traffic entering/exiting the mobile core network, respectively. For the case of MBMS, the home gateway **112** terminates the IP multicast request and controls the flow of the IP multicast traffic in the core network.

(11) The UE **102** uses RAT1 to transport the necessary control plane (CP) data. The CP data includes radio and core network signaling data, service registration data, and security-related signaling that is used to deliver the necessary decryption keys from the Broadcasting/Multicasting Service Center (BM-SC) **114** to the UE **102**.

(12) The BM-SC **114** is the network entity that manages the functions of subscription management, user authentication, key distribution, and multimedia content delivery. The BM-SC **114** provides

functions for broadcasting user service provisioning and delivery. It serves as an entry point for broadcasting traffic transmissions, and authorizes and initiates the establishment of broadcasting traffic transport bearers. It also distributes the service announcements that schedule the delivery of a broadcasting service.

(13) The home gateway **112** directly transmits the broadcasting/multicasting user plane (UP) traffic to the RAT2 RC **108** via controlling nodes, thereby bypassing the mobility anchor **110**.

(14) This approach removes the signaling and processing load from the nodes involved in the delivery of broadcast multimedia traffic, and leaves the capacity of the most widely deployed RAT (in this example, RAT1) unaffected.

(15) UE mobility between the two RATs is supported over an interface **116** between the RAT1 RC **106** and RAT2 RC **108**. The coverage of RAT1 may be larger than that of RAT2. Thus, when UE **102** is no longer in the coverage area of RAT2, a request to handover the multimedia transmission is made to RAT1 using messages communicated between the UE **102** and RAT1 RC **106** and RAT2 RC **108**. The handover messages are similar to well-known conventional handover messages [14], but contain modified handover parameters. The UE **102** informs the RAT1 RC **106** that there is a loss of service over RAT2. The message may include additional parameters such as a cause value indicating that the handover is initiated due to the loss of MBMS reception over RAT2, and the MBMS service ID (or MBMS service IDs if more than one service is activated). The RAT1 RC **106** communicates with RAT2 RC **108** (over an interface between the two RCs **116**) to acquire the service context for the particular service that the UE **102** has activated. After establishing the service context at RAT1 RC **106**, the RAT1 RC **106** sends the radio channel setup information to the UE **102** in order to enable the delivery of the service over RAT1.

(16) FIG. **2** illustrates an exemplary embodiment of signaling flow between elements in a mobile network to direct a user plane tunnel for the delivery of multicasting traffic. The following process follows the establishment of a point-to-point packet pipe by the UE **102**. The messages between elements in steps 1-4 are “application layer” signaling messages.

(17) In step 1, the UE **102** receives the service announcement message transmitted by the BM-SC **114**. The service announcement message indicates the details of the broadcasting/multicasting service.

(18) In step 2, the UE **102** communicates signaling messages with the BM-SC **114** over a first wireless network, RAT1, to perform the service registration request.

(19) In step 3, the UE **102** communicates with the BM-SC **114** via signaling messages to request and receive the necessary security keys via RAT1. The security keys are needed to decrypt the received content.

(20) In step 4, the UE **102** communicates signaling messages to join the multicasting service. The messages are intercepted by the home gateway (HGW) **112**.

(21) In step 5, the home gateway **112** requests and receives authorization from the BM-SC **114** for the UE **102** to allow access to the multicasting service.

(22) In step 6, after the accepted authorization is received from the BM-SC **114** at the home gateway **112**, the home gateway **112** proceeds to receive and establish a point-to multipoint program called a point-to-multipoint packet pipe **202** over RAT2.

(23) In step 7, after the point-to-multipoint pipe **202** is established in step 6, the home gateway **112** updates the termination point of the point-to-multipoint pipe **202** in order to bypass the RAT1 Mobility Anchor (MA) **110** from the path of the user plane traffic. The point-to-multipoint service, called the point-to-multipoint packet pipe **202**, transports the multicasting multimedia traffic via the second network, and terminates at the RAT2 RC **108**.

(24) After the establishment of the point-to-multipoint packet pipe **202** over RAT2, the multicasting multimedia traffic is directly transmitted from the home gateway **112** to the RAT2 RC **108**. From the RAT2 RC **108**, the multicasting multimedia traffic is transmitted to the UE **102**.

(25) The above embodiments can be implemented in the Rel.6 3GPP network architecture with

some enhancements. For example, using two different RATs, W-CDMA radio network controllers may be used to transport the MBMS signaling information. W-CDMA is FDD-based. At the same time, TD-CDMA RNCs may be used in the downlink-only mode to transport the MBMS multimedia traffic. TD-CDMA is TDD-based. In another embodiment, the two RATs may be W-CDMA RNCs. In another embodiment, the two RATs may be TD-CDMA RNCs.
(26) FIG. **3** illustrates an embodiment using UMTS technology. The UE **302** communicates with a W-CDMA radio network controller (RNC) **306**, which defines a first network, and a TD-CDMA RNC **308**, which defines a second network, via node Bs **304**. The first network communicates uplink and downlink control signals to the UE **302**. The second network transmits a point-to-multipoint signal to the UE **302** in response to the uplink and downlink control signals between the first network and the UE **302**. (Node B **304** is the base station for the UMTS cellular system.)
(27) The mobility anchor for the UMTS core network (W-CDMA) is the SGSN **310**. The interface, lu-PS **320**, links the W-CDMA RNC **306** with an SGSN **310**.
(28) The home gateway for the UMTS core network is the GGSN **312**. The GGSN **312** couples the point-to-multipoint service center, the BM-SC **314**, with the first and second wireless networks.
(29) The interface, Gn, links the SGSN **310** and GGSN **312** core network nodes. The Gn is separated into Gn-C **324** and Gn-U **326**: Gn-C **324** indicates the transport of the control plane messages, such as those referring to the user plane tunnel establishment and mobility messages; and Gn-U **326** indicates the user plane traffic transport. The Gn interface (Gn-C **324** and Gn-U **326**), in prior art, connect the same nodes (e.g., same RNC and SGSN). In the embodiments of the invention, however, the Gn-C **324** terminates in the W-CDMA RNC **306**, compared to the Gn-U **326**, which terminates in the TD-CDMA RNC **308**.
(30) Two interfaces exist between the GGSN **312** and the BM-SC **314**. The Gmb **328** is a MBMS-specific interface. It is used for signaling message exchanges between GGSN **312** and BM-SC **314**. A signaling message exchange may include user-specific signaling (such as user-specific charging messages) and MBMS-specific signaling messages (such as the GGSN registration in a BM-SC). The Gi interface **330** is located between the GGSN **312** and the external public packet data network. In the embodiments of the invention, Gi **330** connects the BM-SC **314** and the GGSN **312**. Gi **330** transports the UP multimedia traffic to the GGSN **312** designated to receive the MBMS user service.
(31) The lur **316** interface is located between two RNCs. In an exemplary embodiment, the lur **316** may connect two RNCs of the same RAT (e.g., W-CDMA or TD-CDMA). In other embodiments of the invention, the lur **316** may connect two RNCs of different RATS, for example, a W-CDMA RNC and a TD-CDMA RNC. The lur interface **316** may transfer a downlink channel setup request from the first RNC to a second RNC, and a downlink channel acknowledgment from the second RNC to the first RNC. An embodiment of a specific signaling exchange between the RNCs is shown in the signaling flow of FIG. **4** and FIG. **5**.
(32) If the MBMS service embodiment uses the broadcast delivery mode as defined in [1], the UE **302** uses the W-CDMA RNC **306** to perform service registration, authenticate with the BM-SC **314**, obtain the necessary encryption keys as defined in [4], and receive the service announcement. Next, the UE **302** performs the necessary procedures to "tune in" to the appropriate MBMS traffic channel (MTCH) transmitted over the UMTS TD-CDMA air-interface in order to receive the MBMS multimedia traffic and then use the previously received keys to decrypt the traffic.
(33) However, if the particular service uses the multicast delivery mode, then the UE may perform multicast delivery mode signaling procedures, described in [1], [4], and [5] with modifications explained in the detailed description of FIG. **4**.
(34) FIG. **4** illustrates an example of a signaling message flow that is implemented when the MBMS multicast delivery mode signaling procedures uses a "one tunnel" approach and a downlink-only TD-CDMA radio interface.
(35) In steps 1-4, the current MBMS signaling procedures apply as described in the relevant 3GPP

specifications [1], using the messages defined in [4] and [2], utilizing the default packet data protocol (PDP) context over W-CDMA bearers.

(36) In step 1, the service announcement messages communicated between the UE **302** and BM-SC **314** indicate that the transport of the MBMS traffic will take place over the TD-CDMA network.

(37) In step 2, the service registration request messages are communicated between the BM-SC **314** and the UE **302**.

(38) Step 3 communicates the modulation request messages between the UE **302** and the BM-SC **314**.

(39) In step 4, the UE **302** communicates messages to the BM-SC **314** indicating that it wants to receive messages addressed to a specific multicast group.

(40) For steps 5-9, the current MBMS signaling procedures apply as described in the relevant 3GPP specifications [1], using the messages defined in [9], [10], and [11].

(41) In steps 5-6, signaling messages between the BM-SC **314** and the GGSN **312** consist of a service authorization request from the GGSN **312** and a service authorization answer from the BM-SC **314** (AA request/answer).

(42) In step 7, after receiving an authorization answer, the GGSN **312** sends a MBMS notification request message to the SGSN **310**. In step 8, the SGSN **310** then communicates a "Request MBMS Context Activation" message to the UE **302**. In step 9, the UE **302** communicates an "Activate MBMS Context Request" to the SGSN **310**.

(43) The "one tunnel approach" described in and indicates the reserved "not allocated" value for the traffic path as the lu bearer is not yet allocated in the RAN.

(44) In step 10, the message "Create MBMS Context Request" is sent from the SGSN **310** to the GGSN **312**.

(45) For steps 11-13, the current MBMS signaling procedures apply as described in the relevant 3GPP specifications [1], using the messages defined in [9] and [10].

(46) In steps 11-12, signaling messages between the BM-SC **314** and the GGSN **312** includes a service authorization request from the GGSN **312**, and a service authorization answer from the BM-SC **314** (AA request/answer).

(47) In step 13, a "Create MBMS context response" message is sent from the GGSN **312** to the SGSN **310**.

(48) In step 14, the SGSN **310** allocates the appropriate resources in the RAN for the MBMS context by exchanging signaling information with the W-CDMA RNC **306** and the TD-CDMA RNC **308**, separating the CP and UP paths over the lu-PS interface **320** and the lur interface **316** by a downlink channel setup request.

(49) FIG. **5** illustrates in more detail the MBMS context provisioning in the Radio Access Network (RAN), shown in step 14 of FIG. **4**.

(50) In step 14a, the Radio Access Bearer (RAB) path setup request is sent to the W-CDMA RNC **306** from the SGSN **310**. This request includes a Tunnel Endpoint ID (TEID) of the GGSN **312** and the address of the GGSN **312**.

(51) In step 14b, the W-CDMA RNC **306** communicates the TEID and the address of the GGSN **312** to the corresponding TD-CDMA RNC **308** over an "lur-like" interface. Also, the TD-CDMA RNC **308** is requested for RAB establishment over the TD-CDMA network by a "Radio Access Bearer Path Setup request".

(52) In step 14c the TD-CDMA RNC **308** reserves the resources for RAB establishment and communicates to the W-CDMA RNC **306** indicating a Tunnel Endpoint ID (TEID) at the TD-CDMA RNC **308** for the RAB and the address of the TD-CDMA RNC **308** in a "Radio Access Bearer Path Setup response" message.

(53) In step 14d, the information on TEID at TD-CDMA RNC **308** and the address of the TD-CDMA RNC **308** is delivered to the SGSN **310** from the W-CDMA RNC **306** by a "Radio Access Bearer Path Setup response" message.

(54) In step 15, the current MBMS signaling procedures apply as described in the relevant 3GPP specification [1], using the messages defined in [9], [10], and [11]. The SGSN **310** communicates an “Update MBMS Context” signaling message to the GGSN **312**.
(55) In step 16, after the MBMS Context is provisioned in the RAN, the SGSN **310** receives an “Update MBMS Context Accept” signaling message from the GGSN **312** and updates the MBMS context indicating the RAN address and the TEID of the TD-CDMA RNC **308** to the GGSN **312**.
(56) The GGSN **312** updates the MBMS Context and returns the “Update MBMS Context Response” message. In step 17, the tunnel between the GGSN **312** and the TD-CDMA RNC **308** is established.
(57) In embodiments of the invention, the GGSN **312** sends the MBMS user plane traffic directly to the TD-CDMA RNC **308** following the “one tunnel” approach defined in 3GPP. In this embodiment, the method indicates the MBMS transport mechanism using an element in the MBMS service announcement format.
(58) While the invention has been described in terms of particular embodiments and illustrative figures, those of ordinary skill in the art will recognize that the invention is not limited to the embodiments or figures described. Although embodiments of the present invention are described, in some instances, using UMTS terminology, those skilled in the art will recognize that such terms are also used in a generic sense herein, and that the present invention is not limited to such systems.
(59) Those skilled in the art will recognize that the operations of the various embodiments may be implemented using hardware, software, firmware, or combinations thereof, as appropriate. For example, some processes can be carried out using processors or other digital circuitry under the control of software, firmware, or hard-wired logic. (The term “logic” herein refers to fixed hardware, programmable logic and/or an appropriate combination thereof, as would be recognized by one skilled in the art to carry out the recited functions.) Software and firmware can be stored on computer-readable media. Some other processes can be implemented using analog circuitry, as is well known to one of ordinary skill in the art. Additionally, memory or other storage, as well as communication components, may be employed in embodiments of the invention.
(60) FIG. **6** illustrates a typical computing system **600** that may be employed to implement processing functionality in embodiments of the invention. Computing systems of this type may be used in the BM-SC, the radio controllers, the base stations and the UEs, for example. Those skilled in the relevant art will also recognize how to implement the invention using other computer systems or architectures. Computing system **600** may represent, for example, a desktop, laptop or notebook computer, hand-held computing device (PDA, cell phone, palmtop, etc.), mainframe, server, client, or any other type of special or general purpose computing device as may be desirable or appropriate for a given application or environment. Computing system **600** can include one or more processors, such as a processor **604**. Processor **604** can be implemented using a general or special purpose processing engine such as, for example, a microprocessor, microcontroller or other control logic. In this example, processor **604** is connected to a bus **602** or other communication medium.
(61) Computing system **600** can also include a main memory **608**, such as random access memory (RAM) or other dynamic memory, for storing information and instructions to be executed by processor **604**. Main memory **608** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **604**. Computing system **600** may likewise include a read only memory (“ROM”) or other static storage device coupled to bus **602** for storing static information and instructions for processor **604**.
(62) The computing system **600** may also include information storage system **610**, which may include, for example, a media drive **612** and a removable storage interface **620**. The media drive **612** may include a drive or other mechanism to support fixed or removable storage media, such as a hard disk drive, a floppy disk drive, a magnetic tape drive, an optical disk drive, a CD or DVD drive (R or RW), or other removable or fixed media drive. Storage media **618**, may include, for

example, a hard disk, floppy disk, magnetic tape, optical disk, CD or DVD, or other fixed or removable medium that is read by and written to by media drive **612**. As these examples illustrate, the storage media **618** may include a computer-readable storage medium having stored therein particular computer software or data.

(63) In alternative embodiments, information storage system **610** may include other similar components for allowing computer programs or other instructions or data to be loaded into computing system **600**. Such components may include, for example, a removable storage unit **622** and an interface **620**, such as a program cartridge and cartridge interface, a removable memory (for example, a flash memory or other removable memory module) and memory slot, and other removable storage units **622** and interfaces **620** that allow software and data to be transferred from the removable storage unit **622** to computing system **600**.

(64) Computing system **600** can also include a communications interface **624**. Communications interface **624** can be used to allow software and data to be transferred between computing system **600** and external devices. Examples of communications interface **624** can include a modem, a network interface (such as an Ethernet or other NIC card), a communications port (such as for example, a USB port), a PCMCIA slot and card, etc. Software and data transferred via communications interface **624** are in the form of signals which can be electronic, electromagnetic, optical or other signals capable of being received by communications interface **624**. These signals are provided to communications interface **624** via a channel **628**. This channel **628** may carry signals and may be implemented using a wireless medium, wire or cable, fiber optics, or other communications medium. Some examples of a channel include a phone line, a cellular phone link, an RF link, a network interface, a local or wide area network, and other communications channels.

(65) In this document, the terms “computer program product,” “computer-readable medium” and the like may be used generally to refer to media such as, for example, memory **608**, storage media **618**, or storage unit **622**. These and other forms of computer-readable media may store one or more instructions for use by processor **604**, to cause the processor to perform specified operations. Such instructions, generally referred to as “computer program code” (which may be grouped in the form of computer programs or other groupings), when executed, enable the computing system **600** to perform functions of embodiments of the present invention. Note that the code may directly cause the processor to perform specified operations, be compiled to do so, and/or be combined with other software, hardware, and/or firmware elements (e.g., libraries for performing standard functions) to do so.

(66) In an embodiment where the elements are implemented using software, the software may be stored in a computer-readable medium and loaded into computing system **600** using, for example, removable storage drive **622**, drive **612** or communications interface **624**. The control logic (in this example, software instructions or computer program code), when executed by the processor **604**, causes the processor **604** to perform the functions of the invention as described herein.

(67) It will be appreciated that, for clarity purposes, the above description has described embodiments of the invention with reference to different functional units and processors. However, it will be apparent that any suitable distribution of functionality between different functional units, processors or domains may be used without detracting from the invention. For example, functionality illustrated to be performed by separate processors or controllers may be performed by the same processor or controller. Hence, references to specific functional units are only to be seen as references to suitable means for providing the described functionality, rather than indicative of a strict logical or physical structure or organization.

(68) Although the present invention has been described in connection with some embodiments, it is not intended to be limited to the specific form set forth herein. Rather, the scope of the present invention is limited only by the claims. Additionally, although a feature may appear to be described in connection with particular embodiments, one skilled in the art would recognize that various features of the described embodiments may be combined in accordance with the invention.

(69) Furthermore, although individually listed, a plurality of means, elements or method steps may be implemented by, for example, a single unit or processor. Additionally, although individual features may be included in different claims, these may possibly be advantageously combined, and the inclusion in different claims does not imply that a combination of features is not feasible and/or advantageous. Also, the inclusion of a feature in one category of claims does not imply a limitation to this category, but rather the feature may be equally applicable to other claim categories, as appropriate.
(70) All patents, applications, published applications and other publications referred to herein are incorporated by reference herein in their entirety, including the following references: [1]. 3GPP TS 23.246, "Multimedia/Broadcast Multicast Service (MBMS) User Services; Stage 1", Release 6 [2]. 3GPP TS 26.346, "Multimedia/Broadcast Multicast Service (MBMS); Protocols and codecs", Release 6 [3]. 3GPP TR 29.846, "Multimedia/Broadcast Multicast Service (MBMS); CN1 procedures", Release 6 [4]. 3GPP TS 33.246, "Security of Multimedia Broadcast/Multicast Service", Release 6 [5]. 3GPP TS 25.346, "Introduction of the Multimedia Broadcast/Multicast Service (MBMS) in the Radio Access Network (RAN); Stage 2", Release 6 [6]. Internet Group Management Protocol, IGMPv2, http://www.ietf.org/rfc/rfc2236.txt [7]. "Multicast Listener Discovery (MLD) for IPV6", http://www.ietf.org/rfc/rfc2710.txt [8]. 3GPP TS 32.240, "Charging management; Charging architecture and principles", Release 6 [9]. 3GPP TS 24.008, "Mobile radio interface Layer 3 specification; Core network protocols; Stage 3", Release 6 [10]. 3GPP TS 29.060, "General Packet Radio Service (GPRS); GPRS Tunnelling Protocol (GTP) across the Gn and Gp interface", Release 6 [11]. 3GPP TS 25.331, "Radio Resource Control (RRC); Protocol specification", Release 6 [12]. 3GPP TS 23.809, "One Tunnel solution for Optimisation of Packet Data Traffic", Release 6 [13]. 3GPP TR 23.837, "Feasibility study for transport and control separation in the PS CN domain", Release 4 [14] 3GPP TS 23.060, "Technical Specification Group Services and System Aspects, General Packet Radio Service (GPRS) Service Description, Stage 2, v6.13.0, 2006-06

Claims

1. A user equipment (UE) comprising: a receiver; a transmitter; and a processor, operatively coupled to the receiver and the transmitter, wherein: the transmitter and the processor are configured to communicate, using a first radio access technology (RAT) having a first paired spectrum, wherein the first paired spectrum has a first frequency for downlink and a second frequency for uplink, the transmitter and the processor are configured to transmit, using the first radio access technology (RAT), a service request message including information associated with at least one UE service, the receiver and the processor are configured to receive, based on the transmitted service request message, a message with information to setup a bearer for a second RAT while maintaining the connection with the first RAT, wherein the second RAT is a different RAT than the first RAT, and the receiver and the processor are configured to receive multimedia data using the second RAT and simultaneously receive data using the first RAT.
2. The UE of claim 1 wherein the data is control data.
3. The UE of claim 1 wherein the received multimedia data is received using the bearer for the second RAT.
4. The UE of claim 1 wherein transmissions, using the first RAT and the second RAT, have a same packet data protocol.
5. The UE of claim 4 wherein transmissions, using the first RAT and the second RAT, have same security keys.
6. The UE of claim 1 wherein communication using the first RAT uses frequency division duplexing and communication using the second RAT uses time division duplexing.
7. The UE of claim 1 wherein a coverage area of a base station that the UE is communicating with

using the first RAT is larger than a coverage area of a base station that the UE is communicating with using the second RAT.

8. The UE of claim 1 wherein the transmitter and the processor, in response to determining a loss of service of the second RAT, are configured to inform the first RAT of the loss of service of the second RAT.

9. The UE of claim 1 wherein control plane data is received using the first RAT and user plane data is received using the second RAT.

10. The UE of claim 1 wherein the receiver and the processor are configured to receive signaling using the first RAT and the second RAT from a same mobility control node.

11. A method performed by a user equipment (UE), the method comprising: communicating, using a first radio access technology (RAT) having a first paired spectrum, wherein the first paired spectrum has a first frequency for downlink and a second frequency for uplink, transmitting, using the first radio access technology (RAT), a service request message including information associated with at least one UE service, receiving, based on the transmitted service request message, a message with information to setup a bearer for a second RAT while maintaining the connection with the first RAT, wherein the second RAT is a different RAT than the first RAT, and receiving multimedia data using the second RAT and simultaneously receiving data using the first RAT.

12. The method of claim 11 wherein the simultaneously received data is control data.

13. The method of claim 11 wherein the received multimedia data is received using the bearer for the second RAT.

14. The method of claim 11 wherein transmissions, using the first RAT and the second RAT, have a same packet data protocol.

15. The method of claim 14 wherein transmissions, using the first RAT and the second RAT, have same security keys.

16. The method of claim 11 wherein communication using the first RAT uses frequency division duplexing and communication using the second RAT uses time division duplexing.

17. The method of claim 11 wherein a coverage area of a base station that the UE is communicating with using the first RAT is larger than a coverage area of a base station that the UE is communicating with using the second RAT.

18. The method of claim 11 further comprising in response to determining a loss of service of the second RAT, informing the first RAT of the loss of service of the second RAT.

19. The method of claim 11 wherein control plane data is received using the first RAT and user plane data is received using the second RAT.

20. The method of claim 11 further comprising receiving signaling using the first RAT and the second RAT from a same mobility control node.